GUIDE BEFORE COMMITMENT: WHEN DOES A DIFFUSION MODEL DECIDE WHAT IT DRAWS, AND WHAT DOES CLASSIFIER-FREE GUIDANCE CHANGE?

Anonymous submission

ABSTRACT

Diffusion models often violate part of a compositional prompt. We ask when, during denoising, each constraint type stops depending on the remaining randomness, and what classifier-free guidance (CFG) changes. We re-noise saved states of Stable Diffusion 1.5 and measure how often the outcome survives, corrected for chance agreement, which gives a commitment step t_{50}. Rules and constants were frozen in a hash-signed protocol before any test image existed; only six of nine test shards exist, so all test results are exploratory. Stronger CFG makes object presence commit earlier (-8.67 of 40 steps, 95% CI $[-16.34, -4.72]$, Holm $p = 0.046$), and cross-attention shifts the same way, but the effect disappears if judge errors are independent across seeds. A guidance schedule built on these timings failed its registered test.

***Index Terms*—** Diffusion models, classifier-free guidance, prompt adherence, cross-attention, text-to-image generation

1. INTRODUCTION

Text-to-image diffusion models often draw a convincing image that breaks part of the prompt: an object is missing, a colour lands on the wrong object, or a count or spatial relation is wrong. Adherence is usually reported as one score per image, but these failures are constraint specific. If different constraints are settled at different moments of sampling, a single score hides that structure, guidance applied late may be wasted, and an early internal signal may forecast some constraints while merely reading out others.

Generation proceeds in stages [1, 2, 3], and guidance has been restricted or scheduled over time [4, 5, 6]. What is missing is a direct, per-constraint measurement of *when* an outcome stops depending on the remaining randomness, and a test of whether that timing is useful. This matters in practice because the guidance scale and its schedule are the simplest inference-time controls for trading adherence against diversity, so knowing when guidance acts shows where it can safely be reduced. Besides CFG we vary a second inference-time mechanism, the guidance schedule (CAG) together with the number of denoising steps (20 vs. 40). We ask four questions, each tied to one hypothesis (H1–H4, Section 4): (i) when does each constraint type (presence, count, colour binding, spatial relation, relative size) become persistent (H1); (ii) what does classifier-free guidance (CFG) [7] change about that timing (H2); (iii) can a guidance schedule built on the measurement help (H3); and (iv) does cross-attention reveal the timing (H4)? The specific prediction of H2 is that stronger CFG commits presence earlier by at least 3 of 40 steps; a positive difference would contradict it, and an interval inside ± 1.5 steps would bound it at zero.

We answer with an interventional measurement on Stable Diffusion 1.5 [8]: saved trajectory states are re-noised and denoised again, and outcome agreement, corrected for chance agreement between independent seeds, gives a persistence curve $C(t)$ and half-point t_{50}. Hypotheses, decision rules and development-split constants were frozen in a hash-signed protocol before any test image existed (an internal freeze, not an external registry). Our contributions are a chance-corrected restart-persistence measurement with a closed-form judge-error stress test, a controlled scene-graph benchmark, Commitment-Aligned Guidance (CAG) tested against matched controls, and an account of which conclusions survive on the test data that exist.

2. RELATED WORK

Stages of generation. Our hypotheses rest on a simple mechanism: CFG scales the conditional direction $\varepsilon_\theta(x, c) - \varepsilon_\theta(x, \varnothing)$ of a deterministic trajectory, so stronger guidance should settle which objects appear earlier, and layout is expected to form before attribute binding, consistent with [1, 2]; critical-window theory [3] predicts feature-specific windows. Switching prompts across denoising intervals separates an object's shape from its details [1], stage-specific embeddings control layout and attributes [2], eDiff-I specialises denoisers by stage [9], and critical-window theory predicts feature-specific time windows [3]. These studies locate stages by editing, conditioning or theory; we measure, per constraint type, how much remaining randomness can still change a judged outcome, using re-noising [10, 11] only as a probe.

Guidance schedules. Limiting guidance to an interval of noise levels improves quality [4], annealing the conditioning restores diversity [5], dynamic CFG selects scales online [6], and step selection finds later conditioning more effective [12]. CAG instead fixes its switch step in advance from a held-out measurement.

Attention and evaluation. Cross-attention maps explain [13] and edit [14] generations, attention objectives reduce object neglect and attribute leakage [15, 16], and early attention has been used to select noise and predict quality [17, 18]. We ask whether attention stabilisation coincides with a given constraint's commitment. Detector- and question-answering benchmarks [19, 20, 21] made compositional failures measurable; we treat the judge as an audited instrument. To our knowledge no prior study combines an interventional per-constraint timing measurement, a test of how CFG moves it, an attention comparison and a measurement-derived schedule with matched controls.

3. METHOD

Restart persistence. We use Stable Diffusion 1.5 with DDIM [22] ($\eta = 0$, $T = 40$ steps, 512^2, fp16, pinned revision) and CFG

$$\hat{\varepsilon}_i = \varepsilon_\theta(x_i, \varnothing) + w\,\big(\varepsilon_\theta(x_i, c) - \varepsilon_\theta(x_i, \varnothing)\big). \tag{1}$$

The latent after $t \in \{6, 12, \ldots, 36\}$ steps is saved. A branch renoises it to an earlier step j with fresh $\varepsilon \sim \mathcal{N}(0, I)$ and denoises to the end at the same w:

$$x'_j = \sqrt{r}\, x_t + \sqrt{1-r}\, \varepsilon, \qquad r = \bar{\alpha}(j)/\bar{\alpha}(t), \tag{2}$$

which has exactly the forward marginal at step j, so a branch is a valid state of the sampler and only the injected noise is new. The probe D8 uses $j = \max(t-8, 0)$, ER30 keeps $r \approx 0.30$, and a full resample from pure noise (FULL) is a validity control. For scene s and constraint type κ, $A_{\kappa s}(t)$ is the share of (base, branch) pairs with equal judged outcomes and $A0_{\kappa s}$ the agreement of independent seeds of the same prompt ($A0 = [o(o-1) + (n-o)(n-o-1)]/[n(n-1)]$ for o successes among n seeds). Persistence is the pooled, chance-corrected ratio

$$C_\kappa(t) = \frac{\sum_s \left[A_{\kappa s}(t) - A0_{\kappa s}\right]}{\sum_s \left[1 - A0_{\kappa s}\right]}, \tag{3}$$

and t_{50} is its interpolated first crossing of 0.5 (censored at the grid ends). Higher CFG lowers diversity and so raises $A0$; we therefore compute $A0$ separately at each w. Intervals come from a scene-level bootstrap (10,000 draws, scenes rather than images, because images of a scene are correlated), and Holm's procedure controls the family-wise error over four hypotheses. Chance correction is needed because raw agreement is inflated whenever seeds already agree; four seeds per prompt give each scene its own $A0$ within our compute budget (Stable Diffusion 1.5 is open, widely studied and fits free T4 GPUs).

Attention. Per-step cross-attention maps of each constraint word are averaged over heads and layers (16×16 and 32×32 groups). A type-specific signal's cosine velocity gives t_{stab}, the step at which half of its total path is travelled. Post hoc, we also use localisation (mass in the top 10% of cells) and overlap with the other noun.

CAG. For a prompt with constraint types K,

$$s^\star = \min\big(\lceil \max_{\kappa \in K} t^\kappa_{50,\text{DEV}} \rceil + 2,\ T\big), \tag{4}$$

and guidance is $w_{hi} = 7.5$ before step $s^\star$ and $w_{lo} = 3$ afterwards ($w_{lo} = 1$ failed a pre-declared HPS v2 quality guard, -0.518 vs. a -0.5 limit). This gives $s^\star = 12$ for presence, binding and spatial prompts and 21 for counts. The arms are constant guidance 7.5 (A1), CAG (A2), one global switch step (A3), the reversed schedule with equal mean guidance (A4) and constant guidance at CAG's mean (A6). The adherence score F of an image is the mean of its judged outcome families.

Judge-error stress test (post hoc). If each judged outcome flips independently with probability e, then $A^{\text{obs}} = (1-2e)^2 A + c$ with $c = 2e(1-e)$, for branch and seed pairs alike, and

$$C^{(e)}_\kappa(t) = \frac{\sum_s \left[A^{\text{obs}}_{\kappa s} - A0^{\text{obs}}_{\kappa s}\right]}{\sum_s \left[1 - c - A0^{\text{obs}}_{\kappa s}\right]}, \tag{5}$$

which equals Eq. (3) at $e = 0$. Detector errors are usually content dependent, so this is a worst case.

4. EXPERIMENTAL DESIGN

Benchmark. Prompts come from scene graphs: 100 development (DEV) and 200 test scenes split by object pair, each with a canonical prompt, meaning-preserving variants and one-field counterfactuals, plus a colour×size×relation family on 24 object pairs (1,674 prompts). The five constraint types were chosen because each isolates a distinct failure: presence (object neglect), count (numerosity), binding (attribute leakage), spatial relation and size (relational placement and scale). Complexity is defined as the number of constraints in a prompt (2–6), and one-field counterfactuals let a failure be attributed to a single field. Each prompt uses 4 seeds, $w \in \{2, 5, 7.5, 10, 15\}$ and 20 or 40 steps.

Table 1. Judge audit on DEV (one annotator; population-weighted accuracy, stratified bootstrap).

Type	n	Accuracy [95% CI]	≥ 0.85
presence	159	0.856 [0.802, 0.907]	yes
count	108	0.768 [0.685, 0.843]	no
binding	156	0.800 [0.722, 0.871]	no
spatial	109	0.615 [0.523, 0.706]	no
size	175	0.846 [0.794, 0.897]	no

Table 2. Registered outcomes (Holm over four hypotheses). The H3 row is the 90% interval used by its non-inferiority rule; all other intervals are 95%.

	Estimate [95% CI]	Holm p	Outcome
H1	$\Delta t_{50} = +3.59\ [-10.43, 13.06]$	1.00	inconclusive
H2	$-8.67\ [-16.34, -4.72]$	0.046	supported
H3	$\Delta F = -0.016\ [-0.035, 0.003]^\dagger$	1.00	not supported
H4	concordance $0.30\ [0.20, 0.75]$	1.00	rejected

H1 is exploratory by its DEV gate. †A2−A1, 90% interval; non-inferiority fails ($-0.035 < -0.03$).

Judge and audit. The judge combines OWL-ViT detection [23] with CLIP colour classification [24] at pinned revisions and pre-fixed thresholds. A blinded, stratified, population-weighted audit of DEV items by a single annotator, with a pre-declared 0.85 point-accuracy threshold, gave presence 0.856 (95% CI $[0.802, 0.907]$), size 0.846, binding 0.800, count 0.768 and spatial 0.615. Only presence meets the threshold (Table 1), so all other types are exploratory. A planned question-answering judge was not implemented.

Hypotheses. H1: binding and presence differ in t_{50}. H2: $t_{50}(w{=}15) - t_{50}(w{=}2) < 0$ ($|\Delta| \geq 3$ steps, CI excluding 0), because stronger guidance drifts faster towards text-consistent regions and should settle which objects appear earlier. H3: CAG is non-inferior to constant CFG on the adherence score F (90% lower bound > -0.03), more diverse (DINOv2 [25] cosine distance across seeds) and better than A4, on the reasoning that guidance after commitment is wasted and costs diversity. H4: an outcome that is fixed should be carried by attention that has stopped moving, so the ordering of types by t_{stab} matches that by t_{50} (concordance ≥ 0.8; within-type Spearman 95% CIs above 0 for at least two types; rejected if concordance ≤ 0.5). Support requires the rule and Holm $p < 0.05$. The H2 sign and the H1 ordering are in the signed plan; the mechanisms above are design-document rationale. DEV evidence fixed before testing that H1 is exploratory (precision gate failed: half-width $7.94 > 4$ steps) and that H2 can be supported only through presence.

Execution. Jobs ran on Kaggle T4 GPUs from deterministic manifests in three shards per stage (17,002 judged images, 16,974 after de-duplication). The frozen rules were applied unchanged. A prompt-switch probe was run but not analysed.

5. RESULTS

All test results below come from six of nine test shards and are partial and exploratory. **Registered outcomes.** Table 2 shows that only H2

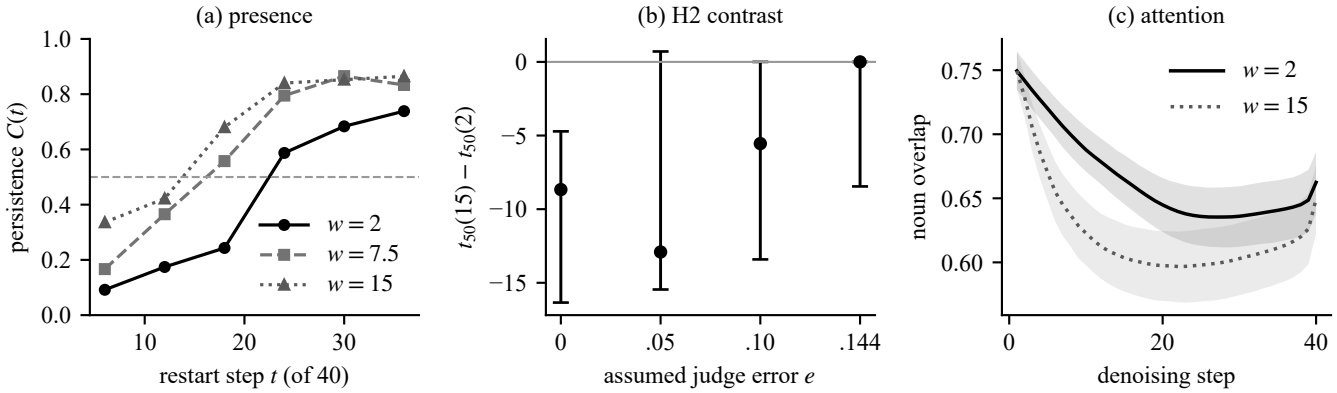


Fig. 1. (a) Presence persistence $C(t)$ at three guidance scales (dashed line: $C = 0.5$). (b) The H2 contrast with 95% scene-bootstrap intervals under an assumed independent judge error e; at $e = 0.144$ (audit) all curves are left-censored and the contrast is not identifiable. (c) Overlap between the two object nouns' cross-attention on identical initial noise (72 scenes with maps at both $w = 2$ and $w = 15$; 221 scenes have maps in total and 159 enter the found-versus-missed pairs; 95% bands). (b) and (c) are post hoc.

passes. H3 fails non-inferiority (90% interval, Table 2; the same estimate has a 95% interval in Table 4) and the comparison with the reversed schedule A4 ($+0.001$, CI $[-0.025, 0.027]$); H4 is rejected by its registered rule. The equal-ratio probe gives presence $t_{50} = 13.3$ at $w = 7.5$ against 16.2 under D8, so absolute commitment steps depend on the probe, as noted in Section 6. A single global switch step (A3) lowers adherence (-0.031, 95% CI $[-0.056, -0.006]$).

CFG and commitment timing. Presence t_{50} falls from 22.5 ($w = 2$) through 16.2 to 13.8 steps ($w = 15$; Fig. 1a). The chance correction does not manufacture this: $A0$ is not higher at $w = 15$ (0.697 vs. 0.729) and raw agreement at $t = 18$ is already higher (0.90 vs. 0.79). Subtracting the DEV full-resample offset (0.159) leaves -10.96 $[-17.95, -3.83]$, and the sign holds in both test shards but reverses in the 10 counting scenes ($+11.4$). Under independent judge errors (Eq. 5) the interval reaches zero at $e = 0.05$, and at the audited $e = 0.144$ every curve is left-censored (Fig. 1b). The result therefore holds only if the presence judge's errors are largely content dependent, which the audit did not measure; the audit images also came from $w = 7.5$ only. Only 67 scenes carry branches at both $w = 2$ and 15 (bootstrap universe 111).

CFG as an inference-time control (post hoc). On the 102 scenes generated at every scale (Table 3), raising w from 2 to 15 increases presence ($0.32 \rightarrow 0.43$), whole-prompt satisfaction ($0.09 \rightarrow 0.17$) and CLIP score, and lowers seed diversity ($0.668 \rightarrow 0.478$), with little change beyond $w \approx 10$. Going from 20 to 40 steps does not change adherence (-0.002, CI $[-0.034, 0.029]$). Whole-prompt satisfaction includes judge types below the audit threshold and falls from 0.33 at two constraints to at most 0.07 at four or more. As effect sizes, from $w = 2$ to 15 presence changes by $+0.11$, whole-prompt satisfaction by $+0.08$ and diversity by -0.19 (differences of the scene means above), and H2's contrast is -8.67 steps, about a fifth of the 40-step trajectory. This is a trade-off, not a confirmed improvement.

Attention. On identical noise, higher CFG lowers noun overlap at step 10 ($0.688 \rightarrow 0.623$, paired -0.065, CI $[-0.081, -0.051]$) and shifts t_{stab} by -9.6 steps for presence (earlier) (CI $[-10.0, -9.2]$; Fig. 1c). Within the same scene, objects the judge later misses have less localised ($+0.062$ for found objects at step 10, CI $[0.049, 0.075]$; 226 pairs) and more overlapping (-0.051, CI $[-0.066, -0.037]$; 216 pairs) noun attention from the first step; this is an association, not a cause. H4 fails because t_{stab} spans only 8.2–14.7 steps across types against 6–20 for t_{50}: the binding signal is the colour word's map, which follows object layout and stabilises early (8.2) although binding persists latest (19.8).

Table 3. CFG sweep on the 102 common scenes (post hoc, 40 steps). t_{50}: presence commitment step; Pres.: presence rate; All: whole-prompt satisfaction; Div.: DINOv2 distance across seeds.

w	t_{50}	Pres.	All	CLIP	Div.
2	22.5	0.32	0.09	29.9	0.668
7.5	16.2	0.41	0.16	32.2	0.514
15	13.8	0.43	0.17	32.5	0.478

Table 4. CAG contrasts with 95% CIs (post hoc; the 90% non-inferiority interval of the registered test is in Table 2). A2: CAG; A6: constant at CAG's mean; A4: reversed. A2−A1 uses 67 scenes, the others 134. Mean guidance: A2 4.64, A1 7.5.

	ΔF	Δ diversity
A2−A1	-0.016 $[-0.039, 0.006]$	$+0.022$ $[0.015, 0.029]$
A2−A6	-0.002 $[-0.026, 0.023]$	-0.013 $[-0.026, -0.001]$
A2−A4	$+0.001$ $[-0.025, 0.027]$	-0.030 $[-0.044, -0.015]$

CAG and transfer from DEV. DEV placed binding at or before the first grid step and presence at 9.3 steps; on test, presence is 16.2 and binding 19.8, so CAG's switch at step 12 comes before binding has settled. Matched-mean controls (Table 4) attribute CAG's diversity gain over constant 7.5 (0.022) to its lower mean guidance: constant guidance at the same mean gains 0.027, and at equal mean CAG is less diverse (-0.013) with no adherence difference (-0.002).

6. DISCUSSION AND LIMITATIONS

What the evidence supports. Under our probe, stronger guidance makes presence settle earlier, and attention maps shift in the same direction. This is the only registered rule that passed, and it passed on partial data. Raising CFG from 2 to 15 is a simple inference-time control that raises measured presence and whole-prompt satisfaction at a clear cost in diversity; we do not claim a confirmed or general

improvement, because the sweep is post hoc and whole-prompt satisfaction includes judge types below the audit threshold. The registered intervention (CAG) did not help.

Why CAG failed. Commitment times from about 100 DEV scenes were imprecise (half-width 7.94 steps) and DEV orderings did not replicate on test, so the frozen switch steps were mis-timed for binding. In addition, the registered comparison with constant 7.5 confounds timing with dose, since every CAG arm applies less guidance on average; matched-mean controls were needed to see this, and we recommend them for any guidance-schedule claim.

Validity of the estimator. The full-resample control failed (maximum $|C| = 0.568$ against 0.15). Its gate could not be passed at this sample size (the simulated null had median 1.0 and 0 of 1,000 runs passed), but a presence offset of 0.159 (null $p = 0.008$) remains unexplained: either the fixed four-seed noise panel behind $A0$ or a base–branch generation difference. Absolute t_{50} values are therefore probe relative, while contrasts between guidance scales are less affected.

Limitations. (i) Three of nine test shards are missing, so the registered confirmatory test has not been run and the comparison with A1 uses 67 scenes. (ii) Only presence passed the judge audit, with one annotator, and the audit images came from $w = 7.5$ only. (iii) The H2 result rests on 67 scenes, reverses for counting, and holds only if judge errors are largely content dependent; a seed-paired audit would settle this. (iv) Post hoc analyses are not multiplicity corrected, and the 88 size-family scenes come from 11 object pairs. (v) Attention maps exist only for canonical 40-step images, so attention results are associations. (vi) We study one model, one sampler and synthetic prompts.

7. CONCLUSION

We contribute (1) a chance-corrected restart-persistence method that measures when each constraint type stops depending on the remaining randomness; (2) evidence that stronger CFG makes object presence commit earlier, with cross-attention shifting the same way (exploratory, and valid only if judge errors are not independent across seeds); and (3) a demonstration that matched-mean controls are needed for any guidance-schedule claim, since CAG's diversity gain over constant guidance was a dose effect and its DEV timings did not transfer. The practical implication is that guidance-schedule claims should be compared against constant guidance at equal mean guidance, and raising CFG trades diversity for adherence rather than improving both. Next steps are the three missing test shards through the frozen pipeline, a seed-paired judge audit and a 400-image check of the full-resample offset.

Answers to the research questions. (1) H2 is supported on the partial test data (-8.67 steps, Holm $p = 0.046$); H1 is inconclusive, H3 is not supported and H4 is rejected. (2) Higher CFG raises presence and whole-prompt satisfaction and lowers diversity, 20 vs. 40 steps changes nothing measurable, and only presence has a validated judge, so per-type claims beyond presence are exploratory. (3) Attention responds to guidance in the same direction as persistence (-9.6 steps in t_{stab} for presence) and shows missed objects early, but it does not reproduce the type ordering. (4) A simple inference-time intervention does not improve adherence for free: raising CFG trades diversity for adherence, and CAG failed its registered test.

Reproducibility. All reported numbers are generated from saved, checksummed outputs by versioned scripts, and re-judging all 16,974 images on different hardware reproduced every judged outcome. Images, latents and raw human labels are not distributed.

AI-assistance disclosure. Code, analysis scripts and manuscript drafts were produced with AI assistance under the author's direction; the DEV audit labels were produced by the author without AI.

8. REFERENCES

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